

1	(a) $v = f\lambda$	C1	
	$\lambda = (3.0 \times 10^8)/(4.6 \times 10^{20})$	C1	
	$(= 6.52 \times 10^{-13} =) 0.65(2)\text{pm}$	A1	[3]
	(b) $t = (8.5 \times 10^{16})/(3.0 \times 10^8)$	C1	
	$(= 2.83 \times 10^8 =) 0.28(3)\text{Gs}$	A1	[2]
	(c) mass, power and temperature all underlined and no others	B1	[1]
	(d) (i) arrow in the direction 30° to 40° south of east	B1	[1]
	(ii) triangle of velocities completed (i.e. correct scale diagram) or correct working given e.g. $[14^2 + 8.0^2 - 2(14)(8.0) \cos 60^\circ]^{1/2}$ or $[(14 - 8.0 \cos 60^\circ)^2 + (8.0 \sin 60^\circ)^2]^{1/2}$	C1	
	resultant velocity = $12(.2)$ (or 12.0 to 12.4 from scale diagram) ms^{-1}	A1	[2]